

Quantifying Moderator Effects on High-Dimensional Multi-Subject Time Series

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Abstract

In the present work, we provide Supplementary Materials to the paper “Quantifying Moderator Effects on High-Dimensional Multi-Subject Time Series.” In Section S1, we provide the proof of our Theorem 1, from Section 6 of our main paper. In Section S2 we provide additional simulation results, comparing the performance of our proposed MOD-VAR with methods in the literature in terms of forecasting, estimation error and GCN recovery.

Keywords: Adaptive estimation, Granger-causal networks, network regression, vector autoregressive models, weighted networks.

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S1 Consistency of MOD-VAR

S1.1 Setting and Assumptions

Consider observations on n subjects, with vector of p moderators $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \in \mathbb{R}^p$ independent and identically distributed random vectors with $\mathbb{E}[\mathbf{X}_k] = \mathbf{0}$ and entrywise bounded support: $|X_{ki}| \leq B$. We consider $\Phi^{(0)*}, \Phi^{(M1)*}, \dots, \Phi^{(Mp)*} \in \mathbb{R}^{d \times d}$ be the true model parameters, and we define $\Phi^* : \mathbb{R}^p \rightarrow \mathbb{R}^{d \times d}$ via the mapping $\Phi^*(\mathbf{x}) = \Phi^{(0)*} + \sum_{r=1}^p x_r \Phi^{(Mr)*}$. For each subject $k \in [n]$, we have T measurements of a d -dimensional time-series $\{\mathbf{Y}_t^{(k)} : 1 \leq t \leq T\}$, with the distribution:

$$\mathbf{Y}_t^{(k)} = \Phi^*(\mathbf{X}_k) \mathbf{Y}_{t-1}^{(k)} + \boldsymbol{\varepsilon}_t^{(k)}, \quad \forall 2 \leq t \leq T, \quad (\text{S1})$$

where $\boldsymbol{\varepsilon}_2^{(k)}, \boldsymbol{\varepsilon}_3^{(k)}, \dots, \boldsymbol{\varepsilon}_T^{(k)} \stackrel{i.i.d.}{\sim} N_d(\mathbf{0}, \Sigma_{\boldsymbol{\varepsilon}}^{(k)})$ for all $k \in [n]$. We denote:

$$\mathbf{Z}^{(k)} = \begin{pmatrix} (\mathbf{Y}_T^{(k)})' \\ (\mathbf{Y}_{T-1}^{(k)})' \\ \vdots \\ (\mathbf{Y}_2^{(k)})' \end{pmatrix}, \quad \mathbf{Y}^{(k)} = \begin{pmatrix} (\mathbf{Y}_{T-1}^{(k)})' \\ (\mathbf{Y}_{T-2}^{(k)})' \\ \vdots \\ (\mathbf{Y}_1^{(k)})' \end{pmatrix}, \quad \mathbf{E}^{(k)} = \begin{pmatrix} (\boldsymbol{\varepsilon}_T^{(k)})' \\ (\boldsymbol{\varepsilon}_{T-1}^{(k)})' \\ \vdots \\ (\boldsymbol{\varepsilon}_2^{(k)})' \end{pmatrix}. \quad (\text{S2})$$

Thus, we can summarize the autoregressive structure for each subject given by (S1) via the matrix form equation:

$$\mathbf{Z}^{(k)} = \mathbf{Y}^{(k)} (\Phi^*(\mathbf{X}_k))' + \mathbf{E}^{(k)}.$$

For all $k \in [n]$, let $\mathbf{X}_k^* = (1, \mathbf{X}_k')' \in \mathbb{R}^{p+1}$ the moderator vector for subject k , adding the zero-indexed intercept term $X_{k0}^* = 1$. We define $\Phi^* := [\Phi^{(0)*}, \Phi^{(M1)*}, \dots, \Phi^{(Mp)*}] \in \mathbb{R}^{d \times (d+dp)}$, and

$$\mathbf{W}^{(k)} = [\mathbf{Y}^{(k)}, X_{k1} \mathbf{Y}^{(k)}, \dots, X_{kp} \mathbf{Y}^{(k)}] = (\mathbf{X}_k^*)' \otimes \mathbf{Y}^{(k)} \in \mathbb{R}^{(T-1) \times (d+pd)}.$$

Then, we can write $\mathbf{Y}^{(k)} (\Phi^*(\mathbf{X}_k))' = \mathbf{W}^{(k)} (\Phi^*)'$. Now, we define

$$\mathbf{Z} = \begin{pmatrix} \mathbf{Z}^{(1)} \\ \mathbf{Z}^{(2)} \\ \vdots \\ \mathbf{Z}^{(n)} \end{pmatrix}, \quad \mathbf{W} = \begin{pmatrix} \mathbf{W}^{(1)} \\ \mathbf{W}^{(2)} \\ \vdots \\ \mathbf{W}^{(n)} \end{pmatrix}, \quad \mathbf{E} = \begin{pmatrix} \mathbf{E}^{(1)} \\ \mathbf{E}^{(2)} \\ \vdots \\ \mathbf{E}^{(n)} \end{pmatrix}. \quad (\text{S3})$$

Then, we summarize the entire model to our simplified matrix notation $\mathbf{Z} = \mathbf{W} (\Phi^*)' + \mathbf{E}$.

For a matrix $\mathbf{A} = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_m] \in \mathbb{R}^{k \times m}$ with column vectors $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^k$, we

denote $\text{vec}(\mathbf{A}) = (\mathbf{a}'_1, \mathbf{a}'_2, \dots, \mathbf{a}'_m)' \in \mathbb{R}^{km}$. Notice that, for any $\Phi \in \mathbb{R}^{d \times (d+dp)}$,

$$\begin{aligned} \|\mathbf{Z} - \mathbf{W}\Phi'\|_F^2 &= \|\text{vec}(\mathbf{Z}) - \text{vec}(\mathbf{W}\Phi')\|_2^2 = \|\text{vec}(\mathbf{Z}) - (\mathbf{I}_d \otimes \mathbf{W})\text{vec}(\Phi')\|_2^2 \\ &= \text{vec}(\mathbf{Z})'\text{vec}(\mathbf{Z}) - 2\text{vec}(\mathbf{Z})'(\mathbf{I}_d \otimes \mathbf{W})\text{vec}(\Phi') + \text{vec}(\Phi')'(\mathbf{I}_d \otimes \mathbf{W}'\mathbf{W})\text{vec}(\Phi'). \end{aligned}$$

To simplify our notation, we define $\boldsymbol{\beta} = \text{vec}(\Phi')$, $\mathbf{y} = \text{vec}(\mathbf{Z})$, $\boldsymbol{\gamma} = \frac{1}{N}(\mathbf{I}_d \otimes \mathbf{W}')\mathbf{y}$ and $\hat{\Gamma} = \frac{1}{N}(\mathbf{I}_d \otimes \mathbf{W}'\mathbf{W})$, $\boldsymbol{\beta}^* = \text{vec}((\Phi^*)')$. Finally, we denote $N = (T-1)n$ and $q = (p+1)d^2$. Then, assuming that $\lambda := \lambda_1 = \lambda_2$, we can denote our proposed optimization problem:

$$\hat{\Phi} = \underset{\Phi \in \mathbb{R}^{d \times (d+dp)}}{\text{argmin}} \frac{1}{N} \|\text{vec}(\mathbf{Z}) - \text{vec}(\mathbf{W}\Phi')\|_2^2 + \lambda \|\Phi\|_1 \quad (\text{S4})$$

into the equivalent form

$$\hat{\boldsymbol{\beta}} = \underset{\boldsymbol{\beta} \in \mathbb{R}^{(p+1)d^2}}{\text{argmin}} -2\boldsymbol{\beta}'\boldsymbol{\gamma} + \boldsymbol{\beta}'\hat{\Gamma}\boldsymbol{\beta} + \lambda \|\boldsymbol{\beta}\|_1. \quad (\text{S5})$$

First, we consider the following set of high-level assumptions.

Definition 1 (Restricted Eigenvalue). We say symmetric matrix $\Gamma \in \mathbb{R}^{q \times q}$ satisfies the restricted eigenvalue condition with parameters $\alpha, \tau > 0$ if for all $\boldsymbol{\theta} \in \mathbb{R}^q$,

$$\boldsymbol{\theta}'\Gamma\boldsymbol{\theta} \geq \alpha \|\boldsymbol{\theta}\|_2^2 - \tau \|\boldsymbol{\theta}\|_1^2. \quad (\text{S6})$$

We denote that a matrix $\Gamma \in \mathbb{R}^{q \times q}$ satisfies (S6) as $\Gamma \in \text{RE}(\alpha, \tau)$.

Assumption 1. *There exists a function $Q(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B) > 0$ such that,*

$$\|\boldsymbol{\gamma} - \hat{\Gamma}\boldsymbol{\beta}^*\|_\infty \leq Q(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B) \sqrt{\frac{2 \log(d) + \log(p)}{N}}.$$

Definition 2 (Conditional spectral Density). Assume $(\mathbf{X}_k, \{\mathbf{Y}_t^{(k)} \in \mathbb{R}^d\}_{t=1}^T)$ is a moderator vector autoregressive model, where $\{\mathbf{Y}_t^{(k)} \in \mathbb{R}^d\}_{t=1}^T$ follows (S1). We define the following.

- We define the *conditional autocovariance function* as $\Gamma_{Y^{(k)}|X_k}(h) = \text{Cov}(\mathbf{Y}_t^{(k)}, \mathbf{Y}_{t+h}^{(k)} | \mathbf{X}_k)$.
- We define the *conditional spectral density function* as

$$f_{Y^{(k)}|X_k}(\theta) := \frac{1}{2\pi} \sum_{\ell=-\infty}^{\infty} \Gamma_{Y^{(k)}|X_k}(\ell) e^{-i\ell\theta}, \quad \theta \in [-\pi, \pi].$$

- For any spectral density function f , we denote the maximum spectral eigenvalue by

$$\mathcal{M}(f) := \sup_{\theta \in [-\pi, \pi]} \Lambda_{\max}(f(\theta)).$$

- For any $\mathbf{x} \in \mathbb{R}^p$, consider the matrix polynomial $\mathcal{A}_{\mathbf{x}}(z) = \mathbf{I}_d - \Phi^*(\mathbf{x})z$, for $z \in \mathbb{C}$.

Then, we define the quantities:

$$\mu_{\min}(\mathcal{A}_{\mathbf{x}}) := \min_{|z|=1} \Lambda_{\min}(\mathcal{A}_{\mathbf{x}}^*(z)\mathcal{A}_{\mathbf{x}}(z)); \quad (\text{S7})$$

$$\mu_{\max}(\mathcal{A}_{\mathbf{x}}) := \max_{|z|=1} \Lambda_{\max}(\mathcal{A}_{\mathbf{x}}^*(z)\mathcal{A}_{\mathbf{x}}(z)). \quad (\text{S8})$$

Definition 3 (α -Orlicz norms). For a random variable $X \in \mathbb{R}$ and $\alpha > 0$, we define the α -Orlicz norm of X as

$$\|X\|_{\psi_{\alpha}} := \inf\{c > 0 : \mathbb{E}[\exp\{|X|^{\alpha}/c^{\alpha}\}] \leq 2\}.$$

S1.2 Useful Lemmas

Lemma 1 (Lemma 12 in Loh and Wainwright [3]). *Let $\Gamma \in \mathbb{R}^{q \times q}$, $s \geq 1$ and $\delta > 0$. Suppose Γ satisfies the deviation bound*

$$\sup_{\|\mathbf{v}\|_2=1, \|\mathbf{v}\|_0 \leq 2s} |\mathbf{v}'\Gamma\mathbf{v}| < \delta.$$

Then, for all $\mathbf{v} \in \mathbb{R}^q$

$$|\mathbf{v}'\Gamma\mathbf{v}| \leq 27\delta(\|\mathbf{v}\|_2^2 + \frac{1}{s}\|\mathbf{v}\|_1^2).$$

Lemma 2 (Lemma B.1 of Basu and Michailidis [1]). *If $\mathbf{W}'\mathbf{W}/N \in \text{RE}(\alpha, \tau)$, then the matrix $I_d \otimes \mathbf{W}'\mathbf{W}/N$ satisfies $I_d \otimes \mathbf{W}'\mathbf{W}/N \in \text{RE}(\alpha, \tau)$ as well.*

Lemma 3 (Proposition 2.4 of Basu and Michailidis [1]). *Consider a centered Gaussian VAR time series $\{Y_t\}_{t \in \mathbb{Z}}$ with VAR coefficient matrix $\Phi \in \mathbb{R}^{d \times d}$ such that $\|\Phi\|_2 < \omega < 1$, and autocovariance function $\{\Gamma_Y(h) \in \mathbb{R}^{d \times d}\}_{h \in \mathbb{Z}}$, noise matrix $\mathbf{E} \in \mathbb{R}^{T \times d}$ such that $\mathcal{M}(f_Y) < \infty$. Let $\mathbf{Y} = [\mathbf{Y}_1, \mathbf{Y}_2, \dots, \mathbf{Y}_T]' \in \mathbb{R}^{T \times d}$ and $\mathbf{E} = [\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2, \dots, \boldsymbol{\varepsilon}_T]' \in \mathbb{R}^{T \times d}$. There exists a constant $c > 0$ such that, for any s -sparse vectors $\mathbf{v}, \mathbf{u} \in \mathbb{R}^d$ with $\|\mathbf{v}\|_2, \|\mathbf{u}\|_2 \leq 1$,*

$$\begin{aligned} \mathbb{P} \left[\left| \frac{[\mathbf{Y}'\mathbf{Y}]_{ij}}{T} - [\Gamma_Y(0)]_{ij} \right| > 3\Lambda_{\max}(\Sigma_{\varepsilon}) \frac{\eta}{(1-\omega)^2} \right] &\leq 6 \exp \left[-cT \min(\eta, \eta^2) \right], \\ \mathbb{P} \left[|\mathbf{u}'(\mathbf{Y}'\mathbf{E}/T)\mathbf{v}| > 2\pi\Lambda_{\max}(\Sigma_{\varepsilon}) \left(1 + \frac{1 + \mu_{\max}(\mathcal{A})}{\mu_{\min}(\mathcal{A})} \right) \eta \right] &\leq 6 \exp \left[-cT \min(\eta, \eta^2) \right]. \end{aligned}$$

Lemma 4 (Exercises 2.24 and 2.44 of Vershynin [4]). *Let $X \in \mathbb{R}$ a random variable with $\|X\|_{\psi_{\alpha}} < \infty$. Then, there exists a constant $c > 0$ such that,*

- $\|X - \mathbb{E}X\|_{\psi_{\alpha}} \leq c\|X\|_{\psi_{\alpha}}$; and,
- if $|X| \leq B$, then $\|X\|_{\psi_{\alpha}} \leq cB$.

Lemma 5 (Theorem 3.4 of Kuchibhotla and Chakraborty [2]). *Suppose that $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n \in \mathbb{R}^q$ are independent, mean-zero vectors such that, for some $\alpha > 0$ and $K_{n,q}, \eta_{n,q} > 0$,*

$$\max_{1 \leq i \leq n} \max_{1 \leq j \leq q} \|X_{ij}\|_{\psi_\alpha} \leq K_{n,q}, \quad \max_{1 \leq j \leq q} \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_{ij}^2] \leq \eta_{n,q}. \quad (\text{S9})$$

Then, for any $t > 0$, with probability at least $1 - 3e^{-t}$,

$$\left\| \frac{1}{n} \sum_{i=1}^n X_i \right\|_\infty \leq 7 \sqrt{\frac{\eta_{n,q}(\log(q) + t)}{n}} + \frac{C_\alpha K_{n,q}(\log(2n))^{1/\alpha}(\log(q) + t)^{1/\alpha^*}}{n}, \quad (\text{S10})$$

where $\alpha^ := \min\{1, \alpha\}$ and $C_\alpha > 0$ is some constant depending only on α .*

Lemma 6 (Proposition D.2 of Kuchibhotla and Chakraborty [2]). *Let $W_1, W_2, \dots, W_k$ are possibly dependent random variables satisfying $\|W_i\|_{\psi_{\alpha_i}} < \infty$ for some $\alpha_i > 0$. Then,*

$$\left\| \prod_{i=1}^k W_i \right\|_{\psi_\beta} \leq \prod_{i=1}^k \|W_i\|_{\psi_{\alpha_i}}, \quad \text{where} \quad \frac{1}{\beta} := \sum_{i=1}^k \frac{1}{\alpha_i}. \quad (\text{S11})$$

S1.3 Proof of Theorem 1

Proposition 1. *Consider the MOD-VAR estimator $\hat{\beta}$ defined by (S5). Assume that $\hat{\Gamma} \in \text{RE}(\alpha, \tau)$ for $s\tau < \alpha/32$. Furthermore, suppose Assumption 1 holds. Finally, assume that*

$$\lambda \geq 4Q(\beta^*, \Sigma_\varepsilon, B) \sqrt{\frac{2\log(d) + \log(p)}{N}}.$$

Then,

$$\|\hat{\beta} - \beta^*\|_2 \leq \frac{3\sqrt{s}\lambda}{\alpha}; \quad \|\hat{\beta} - \beta^*\|_1 \leq \frac{12s\lambda}{\alpha}.$$

Proof. If $\hat{\beta}$ is the minimizer of (S5),

$$-2\hat{\beta}'\gamma + \hat{\beta}'\hat{\Gamma}\hat{\beta} + \lambda\|\hat{\beta}\|_1 \leq -2(\beta^*)'\gamma + (\beta^*)'\hat{\Gamma}\beta^* + \lambda\|\beta^*\|_1.$$

By denoting $\Delta = \hat{\beta} - \beta^*$, by basic manipulation and the triangle inequality,

$$0 \leq \Delta'\hat{\Gamma}\Delta \leq 2\Delta'(\gamma - \hat{\Gamma}\beta^*) + \lambda\|\Delta_S\|_1 - \lambda\|\Delta_{S^c}\|_1. \quad (\text{S12})$$

By Assumption 1 and Hölder's inequality,

$$2\Delta'(\gamma - \hat{\Gamma}\beta^*) \leq 2Q(\beta^*, \Sigma_\varepsilon, B) \sqrt{\frac{2\log(d) + \log(p)}{N}} \|\Delta\|_1 \leq \frac{\lambda}{2} \|\Delta\|_1. \quad (\text{S13})$$

Combining (S12) and (S13), we have the following two conclusions

$$0 \leq \frac{1}{\lambda} \Delta'\hat{\Gamma}\Delta \leq \frac{3}{2} \|\Delta_S\|_1 - \frac{1}{2} \|\Delta_{S^c}\|_1 \leq \frac{3}{2} \|\Delta_S\|_1; \quad \text{and}, \quad \|\Delta\|_1 \leq 4\|\Delta_S\|_1.$$

Note that $\|\Delta\|_1 \leq 4\|\Delta_S\|_1 \leq 4\sqrt{s}\|\Delta_S\|_2 \leq 4\sqrt{s}\|\Delta\|_2$. Since $\hat{\Gamma} \in \text{RE}(\alpha, \tau)$, we have that,

$$\Delta'\hat{\Gamma}\Delta \geq \alpha\|\Delta\|_2^2 - \tau\|\Delta\|_1^2 \geq \alpha\|\Delta\|_2^2 - \frac{\alpha}{2}\|\Delta\|_2^2 = \frac{\alpha}{2}\|\Delta\|_2^2.$$

Then,

$$\frac{\alpha}{2} \|\Delta\|_2^2 \leq \Delta' \hat{\Gamma} \Delta \leq \frac{3\lambda}{2} \|\Delta_S\|_1 \leq \frac{3\sqrt{s}\lambda}{2} \|\Delta\|_2.$$

Thus, we conclude $\|\hat{\beta} - \beta^*\|_2 \leq 3\sqrt{s}\lambda/\alpha$. Furthermore, $\|\hat{\beta} - \beta^*\|_1 \leq 4\sqrt{s}\|\Delta\|_2 \leq 12s\lambda/\alpha$. $\square$

Proposition 2. Assume that $N > 16 \log(pd^2)$, and

$$C_M := \max_{k \in [n]} \Lambda_{\max}(\Sigma_{\varepsilon}^{(k)}) \cdot \sup_{\mathbf{x} \in \mathbb{R}^p} \left(1 + \frac{1 + \mu_{\max}(\mathcal{A}_{\mathbf{x}})}{\mu_{\min}(\mathcal{A}_{\mathbf{x}})} \right) < \infty. \quad (\text{S14})$$

Then, there exists a constant $c > 0$ such that

$$\mathbb{P} \left[\|\gamma - \hat{\Gamma} \beta^*\|_{\infty} \geq 4\pi C_M \max\{1, B\} \sqrt{\frac{2 \log(d) + \log(p)}{N}} \right] \leq 6 \exp\{-3c(2 \log(d) + \log(p))\}.$$

Proof. First, observe that,

$$\begin{aligned} \gamma &= \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}') \text{vec}(\mathbf{Z}) = \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}' \mathbf{W}) \text{vec}(\Phi^{*'}) + \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}') \text{vec}(\mathbf{E}); \\ \hat{\Gamma} \beta^* &= \frac{1}{N} (\mathbf{I}_d \otimes \mathbf{W}' \mathbf{W}) \text{vec}(\Phi^{*'}). \end{aligned}$$

Thus,

$$\|\gamma - \hat{\Gamma} \beta^*\|_{\infty} = \|\text{vec}(\mathbf{W}' \mathbf{E})/N\|_{\infty} = \|\mathbf{W}' \mathbf{E}/N\|_{\max} = \max_{\substack{0 \leq i \leq p; \\ 1 \leq m, \ell \leq d}} \left| \frac{1}{n} \sum_{k=1}^n \frac{1}{T-1} X_{ki}^* (\mathbf{Y}_{\cdot m}^{(k)})' \mathbf{E}_{\cdot \ell}^k \right|.$$

By Lemma 3, there exists $c > 1$ such that for any $\eta > 0$,

$$\mathbb{P} \left[\left| \frac{1}{T-1} (\mathbf{Y}_{\cdot m}^{(k)})' \mathbf{E}_{\cdot \ell}^{(k)} \right| \geq 2\pi C_M \eta \right] \leq 6 \exp \left\{ -c(T-1) \min(\eta, \eta^2) \right\}.$$

Recall that $|X_{ki}^*| \leq \max\{1, B\}$. Using this, and the fact that $\mathbb{E}[\mathbf{X}_{ki}(\mathbf{Y}_{\cdot m}^{(k)})' \mathbf{E}_{\cdot \ell}^{(k)}] = 0$ for all $m, \ell \in [D]$, we aggregate the independent sub-exponential bounds and obtain,

$$\begin{aligned} &\mathbb{P} \left[\left| \frac{1}{n} \sum_{k=1}^n \frac{1}{T-1} X_{ki}^* (\mathbf{Y}_{\cdot m}^{(k)})' \mathbf{E}_{\cdot \ell}^{(k)} \right| \geq t \right] \\ &\leq 6 \exp \left\{ -\min \left[\frac{cNt^2}{4\pi^2 C_M^2 \max\{1, B\}^2}, \frac{cNt}{2\pi C_M \max\{1, B\}} \right] \right\}, \end{aligned}$$

for some adjusted constant $c > 1$. Notice that, if $t^* = 4\pi C_M \max\{1, B\} \sqrt{\log(pd^2)/N}$ and $N > 16 \log(pd^2)$,

$$\frac{cN(t^*)^2}{4\pi^2 C_M^2 \max\{1, B\}^2} = 4c \log(pd^2) < 2c \sqrt{N \log(pd^2)} = \frac{cNt^*}{2\pi C_M \max\{1, B\}}.$$

Thus,

$$\mathbb{P} \left[\left| \frac{1}{n} \sum_{k=1}^n \frac{1}{T-1} X_{ki}^* (\mathbf{Y}_{\cdot m}^{(k)})' \mathbf{E}_{\cdot \ell}^{(k)} \right| \geq 4\pi C_M \max\{1, B\} \sqrt{\frac{\log(pd^2)}{N}} \right] \leq 6 \exp \left\{ -4c \log(pd^2) \right\}.$$

By applying the union bound, we obtain,

$$\begin{aligned}
& \mathbb{P} \left[\max_{\substack{0 \leq i \leq p; \\ 1 \leq m, \ell \leq d}} \left| \frac{1}{N} \sum_{k=1}^n X_{ki}^* (\mathbf{Y}_{\cdot m}^{(k)})' \mathbf{E}_{\cdot \ell}^{(k)} \right| \geq 4\pi C_M \max\{1, B\} \sqrt{\frac{\log(pd^2)}{N}} \right] \\
& \leq 6 \exp \left\{ \log(pd^2) - 4c \log(pd^2) \right\} \\
& \leq 6 \exp \left\{ -3c \log(pd^2) \right\}.
\end{aligned}$$

□

Proposition 3. *Let,*

$$\begin{aligned}
\Lambda_Y &:= \inf_{\mathbf{x}} \Lambda_{\min}(\Gamma_{Y|\mathbf{x}}(0)), \\
\alpha^* &= \min\{1, \Lambda_{\min}(\Sigma_X)\} \cdot \Lambda_Y/2, \\
\tau^* &= 54\delta.
\end{aligned}$$

Then, there exist choices of constants $\eta, K > 0$ such that, for any $t > 0$, and

$$\delta := \delta(t) = 7 \sqrt{\frac{\eta B^2 (\log(pd^2) + t)}{n}} + \frac{K B^2 \log(2n) (\log(pd^2) + t)}{n},$$

the following statement holds: for all $t > 0$ such that $\alpha^ > 54s\delta$, we have that $\hat{\Gamma} = (\mathbf{I}_d \otimes \mathbf{W}'\mathbf{W})/N \in \text{RE}(\alpha^*, \tau^*)$ with probability at least $1 - 3e^{-t}$.*

Proof. First, by Lemma 2, it suffices to show $\mathbf{W}'\mathbf{W}/N \in \text{RE}(\alpha^*, \tau^*)$. We denote $\Sigma_W = \mathbb{E}[\mathbf{W}'\mathbf{W}/N]$. To achieve this, the steps will be as follows: (i) show that $\mathbf{v}'\Sigma_W\mathbf{v} > K\|\mathbf{v}\|_2^2$ for some constant $K > 0$, (ii) use Lemma 5 to provide non-asymptotic bounds for $|\mathbf{v}'(\mathbf{W}'\mathbf{W}/N - \Sigma_W)\mathbf{v}|$, and (iii) apply Lemma 1 to derive the restricted eigenvalue property for $\mathbf{W}'\mathbf{W}/N$ with probability at least $1 - 3e^{-t}$.

First, we note that,

$$\frac{1}{N} \mathbf{W}'\mathbf{W} = \frac{1}{n} \sum_{k=1}^n \left\{ \mathbf{X}_k^* (\mathbf{X}_k^*)' \otimes \left(\frac{1}{T-1} (\mathbf{Y}^{(k)})' \mathbf{Y}^{(k)} \right) \right\}. \quad (\text{S15})$$

Let $\mathbf{V} \in \mathbb{R}^{d \times (p+1)}$ be a matrix with columns $\mathbf{V} = [\mathbf{v}_0, \mathbf{v}_1, \dots, \mathbf{v}_p]$. Let $\mathbf{v} = (\mathbf{v}'_0, \mathbf{v}'_1, \dots, \mathbf{v}'_p)' \in \mathbb{R}^{(p+1)d}$. Then, observe that,

$$\mathbf{W}^{(k)} \mathbf{v} = [(\mathbf{X}_k^*)' \otimes \mathbf{Y}^{(k)}] \mathbf{v} = \sum_{a=0}^p x_{ka}^* \mathbf{Y}^{(k)} \mathbf{v}_a = \mathbf{Y}^{(k)} \mathbf{V} \mathbf{X}_k^*.$$

Using the same notation for $V \in \mathbb{R}^{d \times (p+1)}$ and $\mathbf{v} \in \mathbb{R}^{(p+1)d}$, we have that

$$\begin{aligned}
\mathbb{E} \left[\frac{1}{N} \mathbf{v}' \mathbf{W}' \mathbf{W} \mathbf{v} \right] &= \frac{1}{N} \sum_{k=1}^n \mathbb{E} [(\mathbf{X}_k^*)' V' (\mathbf{Y}^{(k)})' \mathbf{Y}^{(k)} V \mathbf{X}_k^*] \\
&= \mathbb{E} \left\{ \mathbb{E} \left[\frac{1}{n} \sum_{k=1}^n (\mathbf{X}_k^*)' V' \left(\frac{1}{T-1} (\mathbf{Y}^{(k)})' \mathbf{Y}^{(k)} \right) V \mathbf{X}_k^* \middle| \mathbf{X} \right] \right\} \\
&= \mathbb{E} [(\mathbf{X}_1^*)' V' \Gamma_{Y^{(1)}|X_1}(0) V \mathbf{X}_1^*] \\
&\geq \Lambda_Y \mathbb{E} [(\mathbf{X}_1^*)' V' V \mathbf{X}_1^*] \geq \Lambda_Y \Lambda_{\min}(\Sigma_{X^*}) \|\mathbf{v}\|_2^2 \\
&= \Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\} \|\mathbf{v}\|_2^2.
\end{aligned}$$

The change from $\Lambda_{\min}(\Sigma_{X^*})$ to $\min\{1, \Lambda_{\min}(\Sigma_X)\}$ is given by the fact that,

$$\Sigma_{X^*} := \mathbb{E} (\mathbf{X}_k^* (\mathbf{X}_k^*)') = \begin{pmatrix} 1 & \mathbf{0}' \\ \mathbf{0} & \Sigma_X \end{pmatrix},$$

and, from this, $\Lambda_{\min}(\Sigma_{X^*}) = \min\{1, \Lambda_{\min}(\Sigma_X)\}$.

Now, we focus on providing concentration for $\sup_{\|\mathbf{v}\|_0 \leq 2s; \|\mathbf{v}\|=1} |\mathbf{v}' (\mathbf{W}' \mathbf{W} / N - \Sigma_W) \mathbf{v}|$. Notice that the terms $\{(\mathbf{X}_k^* (\mathbf{X}_k^*)') \otimes [(\mathbf{Y}^{(k)})' \mathbf{Y}^{(k)}] / (T-1)\}_{k=1, \dots, n}$ in (S15) are independent. Then, we may apply Lemma 5. To this aim, observe that for any $1 \leq i, j \leq d$ and $0 \leq \ell, m \leq p$, by Lemma 3,

$$\begin{aligned}
\left| \left[\frac{1}{T-1} (\mathbf{X}_k^* (\mathbf{X}_k^*)') \otimes (\mathbf{Y}^{(k)})' \mathbf{Y}^{(k)} \right]_{i\ell, jm} \right| &= \frac{1}{T-1} |X_{k\ell}^* X_{km}^* (\mathbf{Y}_{\cdot i}^{(k)})' \mathbf{Y}_{\cdot j}^{(k)}| \\
&\leq \frac{\max\{1, B^2\}}{T-1} |(\mathbf{Y}_{\cdot i}^{(k)})' \mathbf{Y}_{\cdot j}^{(k)}| \\
&\leq (\max\{1, B^2\}) \Gamma_{Y^{(k)}|X_k}(0)_{ij} + O_p \left(\frac{C_M \max\{1, B^2\}}{T-1} \right) \\
&= O_p(C_M \max\{1, B^2\}).
\end{aligned}$$

On the other hand,

$$\begin{aligned}
&\max_{\substack{1 \leq i, j \leq d \\ 0 \leq \ell, m \leq p}} \frac{1}{n} \sum_{k=1}^n \mathbb{E} \left[\left(\frac{1}{T-1} X_{k\ell}^* X_{km}^* (\mathbf{Y}_{\cdot i}^{(k)})' \mathbf{Y}_{\cdot j}^{(k)} \right)^2 \right] \\
&= \max_{\substack{1 \leq i, j \leq d \\ 0 \leq \ell, m \leq p}} \mathbb{E} \left[\left(\frac{1}{T-1} X_{1\ell}^* X_{1m}^* (\mathbf{Y}_{\cdot i}^{(1)})' \mathbf{Y}_{\cdot j}^{(1)} \right)^2 \right] \\
&\leq \max_{\substack{1 \leq i, j \leq d \\ 0 \leq \ell, m \leq p}} \max\{1, B^4\} \mathbb{E} \left[\left(\frac{1}{T-1} (\mathbf{Y}_{\cdot i}^{(1)})' \mathbf{Y}_{\cdot j}^{(1)} \right)^2 \right] \\
&\leq O_p(C_M \max\{1, B^4\})
\end{aligned}$$

As a consequence, for appropriately chosen constants $K, \eta > 0$ depending on B and C_M ,

by Lemma 5,

$$\left\| \frac{1}{N} \mathbf{W}' \mathbf{W} - \Sigma_W \right\|_{\max} \leq \delta := 7 \sqrt{\frac{\eta(\log(pd^2) + t)}{n}} + \frac{K \log(2n)(\log(pd^2) + t)}{n}$$

with probability at least $1 - 3e^{-t}$. From this, it can easily be shown that,

$$\sup_{\|\mathbf{v}\|_0 \leq 2s; \|\mathbf{v}\|_2=1} \left| \mathbf{v}' \left(\frac{\mathbf{W}' \mathbf{W}}{N} - \Sigma_W \right) \mathbf{v} \right| \leq 2s\delta.$$

Applying Lemma 1, it follows that, for any $\mathbf{v} \in \mathbb{R}^{(p+1)d}$

$$\left| \mathbf{v}' \left(\frac{\mathbf{W}' \mathbf{W}}{N} - \Sigma_W \right) \mathbf{v} \right| \leq 54s\delta \|\mathbf{v}\|_2^2 + 54\delta \|\mathbf{v}\|_1^2.$$

To conclude, observe that for any $\mathbf{v} \in \mathbb{R}^{(p+1)d}$

$$\begin{aligned} \left| \mathbf{v}' \left(\frac{1}{N} \mathbf{W}' \mathbf{W} \right) \mathbf{v} \right| &\geq |\mathbf{v}' \Sigma_W \mathbf{v}| - \left| \mathbf{v}' \left(\frac{\mathbf{W}' \mathbf{W}}{N} - \Sigma_W \right) \mathbf{v} \right| \\ &\geq (\Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\} - 54s\delta) \|\mathbf{v}\|_2^2 - 54\delta \|\mathbf{v}\|_1^2. \end{aligned}$$

Thus, as long as $\Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\}/2 > 54s\delta$, it follows that, $\mathbf{W}' \mathbf{W}/N \in RE(\alpha^*, \tau^*)$ for $\alpha^* = \Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\}/2$ and $\tau^* = 54\delta$ and $\delta > 0$ is as defined above. $\square$

Theorem. Consider the MOD-VAR estimator $\hat{\Phi} \in \mathbb{R}^{d \times (dp+d)}$ defined by (S4). Assume that Φ^* is s -sparse, *i.e.* $\|\Phi^*\|_0 \leq s$, and we have equal sample size T for all n subjects. Let $N = n(T-1)$ and $C_M = \max_{k \in [n]} \Lambda_{\max}(\Sigma_{\varepsilon}^{(k)}) \cdot \sup_{\mathbf{x} \in \mathbb{R}^p} (1 + (1 + \mu_{\max}(\mathcal{A}_{\mathbf{x}}))/\mu_{\min}(\mathcal{A}_{\mathbf{x}}))$. Then, there exist constants $K, \eta, c > 0$ depending on B and C_M such that the following statements hold. Let $t > 0$, and

$$\delta := \delta(t) = 7 \sqrt{\frac{\eta(\log(pd^2) + t)}{n}} + \frac{K \log(2n)(\log(pd^2) + t)}{n},$$

$$\Lambda_Y := \inf_{\mathbf{x}} \Lambda_{\min}(\Gamma_{Y|\mathbf{x}}(0)),$$

$$\alpha^* := \min\{1, \Lambda_{\min}(\Sigma_X)\} \cdot \Lambda_Y/2,$$

$$\tau^* := 54\delta,$$

$$Q(\beta^*, \Sigma_{\varepsilon}, B) := 4 \max\{1, B\} \pi C_M.$$

Whenever the following inequalities are satisfied

$$\alpha^* > 32s\tau^* \quad (\text{S16})$$

$$N > 16 \log(pd^2) \quad (\text{S17})$$

$$C_M < \infty, \quad (\text{S18})$$

$$\lambda \geq 4Q(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B) \sqrt{\frac{2 \log(d) + \log(p)}{N}}. \quad (\text{S19})$$

our MOD-VAR estimator satisfies the consistency

$$\begin{aligned} \|\hat{\Phi} - \Phi^*\|_F &\leq \frac{6\sqrt{s}\lambda}{\Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\}}; \\ \|\hat{\Phi} - \Phi^*\|_1 &\leq \frac{24s\lambda}{\Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\}}, \end{aligned}$$

with probability at least $1 - 3e^{-t} - 6e^{-3c \log(pd^2)}$.

Proof. By (S16), we have $\alpha^* > 32s\tau^* > 54s\delta$. By Proposition 3 we have $\hat{\Gamma} \in \text{RE}(\alpha^*, \tau^*)$ with probability at least $1 - 3e^{-t}$.

By (S17), (S18), and applying Proposition 2 we have that,

$$\|\boldsymbol{\gamma} - \hat{\Gamma}\boldsymbol{\beta}^*\|_\infty < Q(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B) \sqrt{\frac{2 \log(d) + \log(p)}{N}}$$

with probability at least $1 - 6 \exp\{-3c(2 \log(d) + \log(p))\}$. Therefore, Assumption 1 is satisfied.

Furthermore, by (S19),

$$\lambda \geq 4Q(\boldsymbol{\beta}^*, \Sigma_\varepsilon, B) \sqrt{\frac{2 \log(d) + \log(p)}{N}}.$$

Since the restricted eigenvalue condition is satisfied for $\alpha^* > s\tau^*$, and Assumption 1 holds, we have that:

$$\begin{aligned} \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_2 &\leq \frac{3\sqrt{s}\lambda}{\alpha} = \frac{6\sqrt{s}\lambda}{\Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\}}; \\ \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\|_1 &\leq \frac{12s\lambda}{\alpha} = \frac{24s\lambda}{\Lambda_Y \min\{1, \Lambda_{\min}(\Sigma_X)\}}. \end{aligned}$$

□

The version of Theorem 1 shown in the main manuscript results of a simple reformulation of the theorem proven above, and observing that, since $\sup_{\mathbf{x}} \|\Phi(\mathbf{x})\|_2 < \eta < 1$, $\mu_{\max}(\mathcal{A}_{\mathbf{x}}) \leq (1 + \eta)^2$ and $\mu_{\min}(\mathcal{A}_{\mathbf{x}}) \geq (1 - \eta)^2$. Therefore,

$$1 + \frac{1 + \mu_{\max}(\mathcal{A}_{\mathbf{x}})}{\mu_{\min}(\mathcal{A}_{\mathbf{x}})} \leq 1 + \frac{1 + (1 + \eta)^2}{(1 - \eta)^2} = 2 + \frac{1 + 4\eta}{(1 - \eta)^2}.$$

Since this bound applies for all feasible $\mathbf{x}$, we conclude.

S2 Additional Simulation Results

In Section 6 of our main paper, we present numerical simulation results comparing the performance of our proposed MOD-VAR on data where multi-subject heterogeneity is explained by moderator effects, as well as data where heterogeneity consists of a combination of moderator-dependent and subject-specific components. These comparisons focus on the L_1 -norm parameter estimation error, as well as forecasting error. In the present section, we present numerical simulation results for synthetic data where multi-subject heterogeneity is subject-specific, with no moderator components. These results are presented in Section S2.1. Additionally, we provide simulation comparisons of our proposed method in terms of mean true positive rate (TPR), mean false positive rate (FPR), defined as

$$TPR(\hat{\Phi}, \Phi^*) = \frac{1}{n} \sum_{k=1}^n \frac{|\{(i, j) : \hat{\Phi}_{ij}^{(k)} \neq 0, \Phi_{ij}^{(k)*} \neq 0\}|}{\|\Phi^{(k)*}\|_0},$$

$$FPR(\hat{\Phi}, \Phi^*) = \frac{1}{n} \sum_{k=1}^n \frac{|\{(i, j) : \hat{\Phi}_{ij}^{(k)} \neq 0, \Phi_{ij}^{(k)*} = 0\}|}{|\{(i, j) : \Phi_{ij}^{(k)*} = 0\}|}.$$

These simulations are presented in Section S2.2. Finally, we provide comparisons in terms of computational running time in Section S2.3.

S2.1 Results for Heterogeneous Time Series Without Moderator Effects

We explore the performance of our proposed MOD-VAR and methods in the literature on multi-subject time-series data with subject-specific components and no moderator-dependent effects. We visualize the performance in terms of L_1 -norm parameter estimation error for $d = 10$ and $d = 20$ in Figures S1 and S2, respectively. Furthermore, we visualize results in terms of 1-step forecasting error in Figures S3 and S4.

In Figure S1, we observe that our proposed adaptive MOD-VAR attains the lowest mean L_1 -norm parameter estimation error in most simulation scenarios. In particular, the adaptive Multi-VAR method attains comparable results to our MOD-VAR method when 25% of entries are shared across models. The individual VAR method attains the worst performance across most comparisons. Furthermore, in Figure S2, we observe a visible advantage of both our original and adaptive MOD-VAR over other methods in the

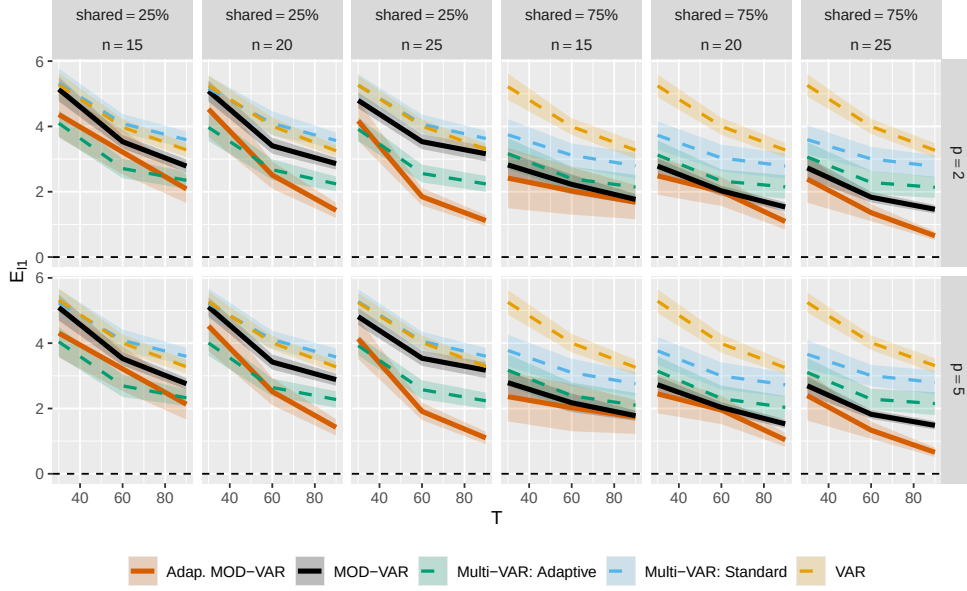


Figure S1: Visualization of the L_1 -norm parameter estimation error of our proposed MOD-VAR and methods in the literature for data where the GCN heterogeneity consists of subject-specific components, with no moderator effects. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

literature. The improved error rate of our proposed MOD-VAR is particularly evident as the number of shared entries increases.

In Figure S3, we compare the 1-step out-of-sample RMSFE accuracy across methods for $d = 10$. Our proposed MOD-VAR shows an improved average performance compared to methods in the literature when the number of shared entries is large. Despite this average improvement, the variation in the estimates is large enough to make a conclusive assessment difficult. In Figure S4, forecasting performance is compared when $d = 20$. Both our original and adaptive MOD-VAR procedures attain the best forecasting performance in most simulation scenarios, closely followed by the individual VAR method.

S2.2 True and False Positive Rates

We provide a comparison of the true positive rate (TPR) and false positive rate (FPR) of our proposed MOD-VAR, as well as methods in the literature in this section. In Figures S5 and S6, we visualize the mean TPR and FPR attained by the compared methods, when the heterogeneity of the models consists only of moderator effects. All compared methods exhibit a trade-off between the TPR and FPR. For example, the adaptive Multi-VAR

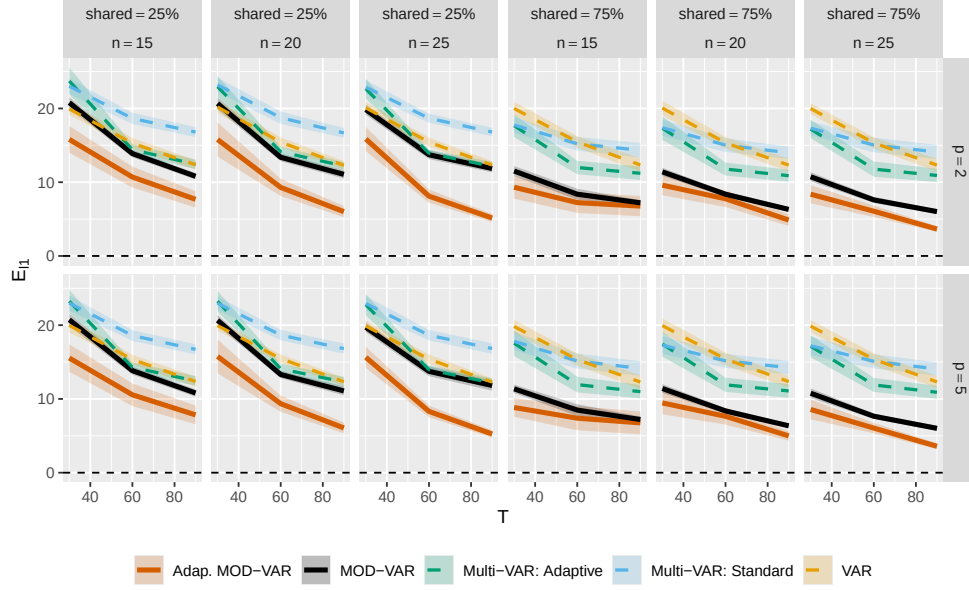


Figure S2: Visualization of the L_1 -norm parameter estimation error of our proposed MOD-VAR and methods in the literature for data where the GCN heterogeneity consists of subject-specific components, with no moderator effects. We explore performance for $d = 20$, varying n , T , p and the proportion of shared/moderated components.

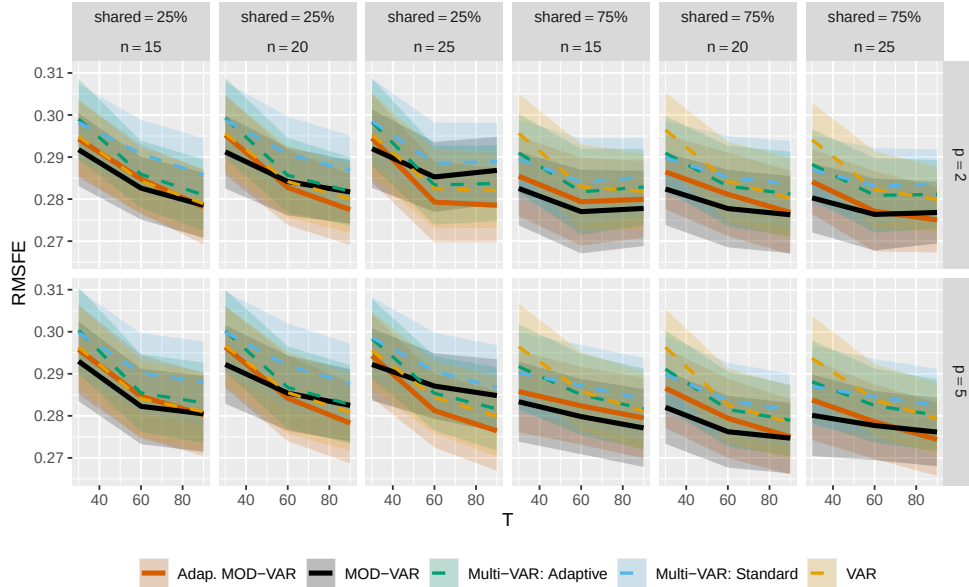


Figure S3: Visualization of the 1-step out-of-sample RMSFE of our proposed MOD-VAR and methods in the literature for data where the GCN heterogeneity consists of subject-specific components, with no moderator effects. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

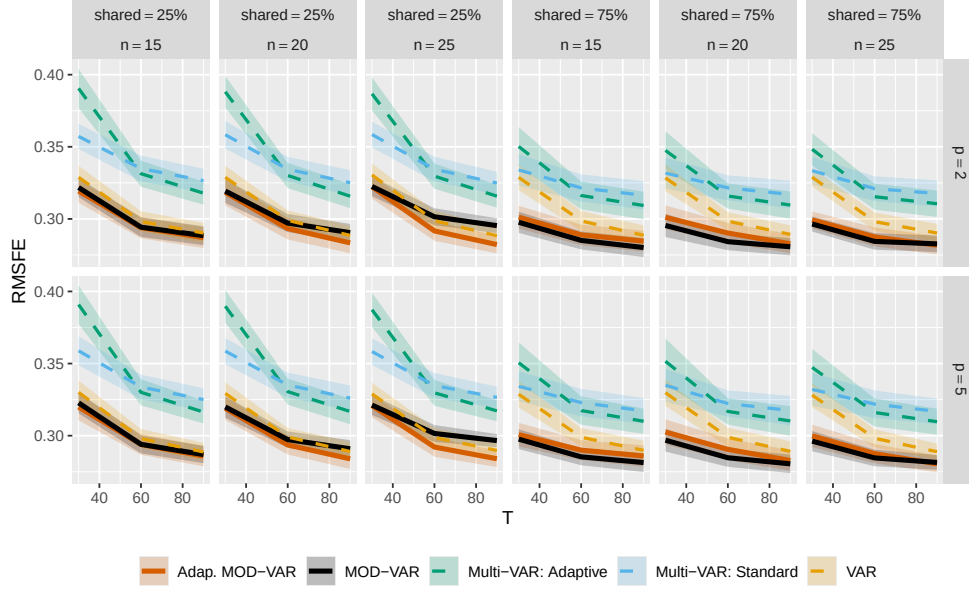


Figure S4: Visualization of the 1-step out-of-sample RMSFE of our proposed MOD-VAR and methods in the literature for data where the GCN heterogeneity consists of subject-specific components, with no moderator effects. We explore performance for $d = 20$, varying n , T , p and the proportion of shared/moderated components.

attains the lowest FPR across all scenarios, at the cost of attaining the lowest TPR across all methods and simulation scenarios. Thus, while the adaptive Multi-VAR procedure tends to select an edge set contained within the true edge set, it tends to have a high false negative rate. On the other hand, the original MOD-VAR has a high TPR, paired with a high FPR across simulation scenarios. The adaptive MOD-VAR attains an interesting balance between TPR and FPR, with the second lowest FPR across all methods and simulation settings, but still enjoys one of the highest TPRs. Thus, we recommend the adaptive MOD-VAR for its reliable pairing of high TPR and low FPR. We draw a similar conclusion for the case where $d = 20$ and heterogeneity is driven only by moderators, as shown in Figures S7 and S8.

We provide additional simulations for data where heterogeneity across subjects is a combination of moderator-dependent and subject-specific components. We compare the mean TPR and FPR when $d = 10$ in Figures S9 and S10, respectively. We provide additional results of TPR and FPR when $d = 20$ in Figures S11 and S12, respectively. Here, we observe a similar pattern: the adaptive MOD-VAR attains one of the highest TPRs, while simultaneously attaining the second lowest FPR. In particular, as shown in both Figure

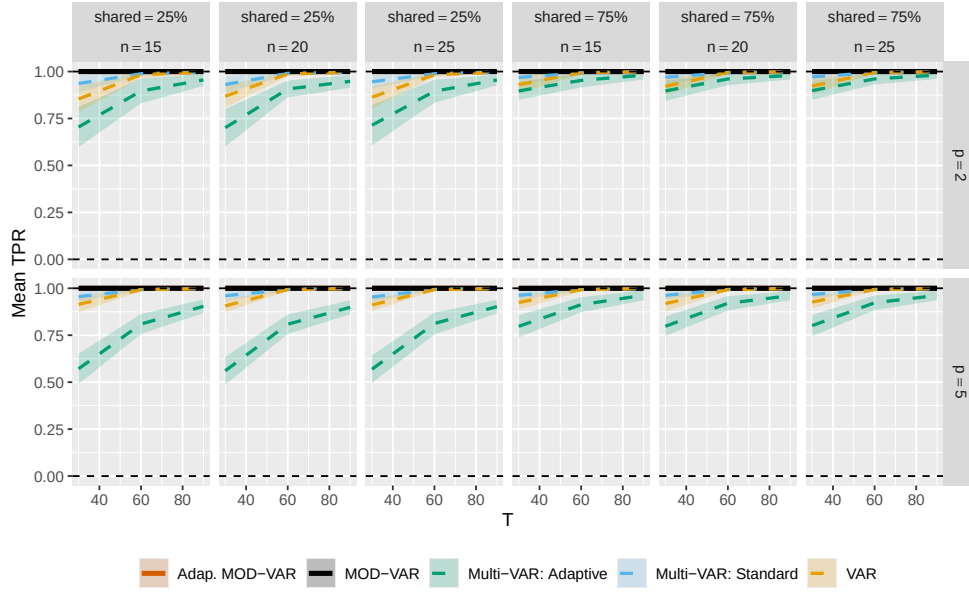


Figure S5: Visualization of the mean true positive rate (TPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects consists of only moderator-dependent components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

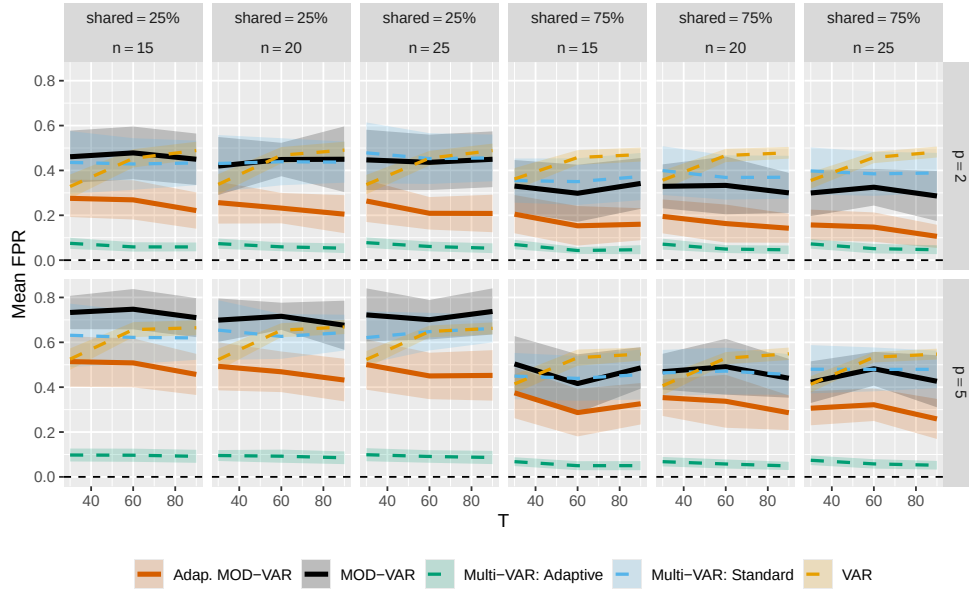


Figure S6: Visualization of mean false positive rate (FPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects consists of only moderator-dependent components, with no subject-specific components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

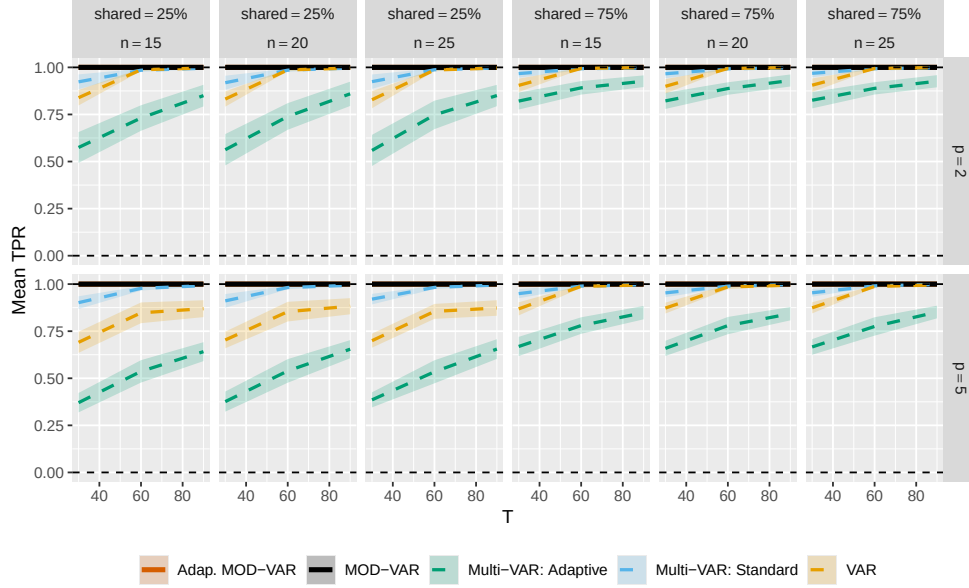


Figure S7: Visualization of the mean true positive rate (TPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects consists of only moderator-dependent components, with no subject-specific components. We explore performance for $d = 20$, varying n , T , p and the proportion of shared/moderated components.

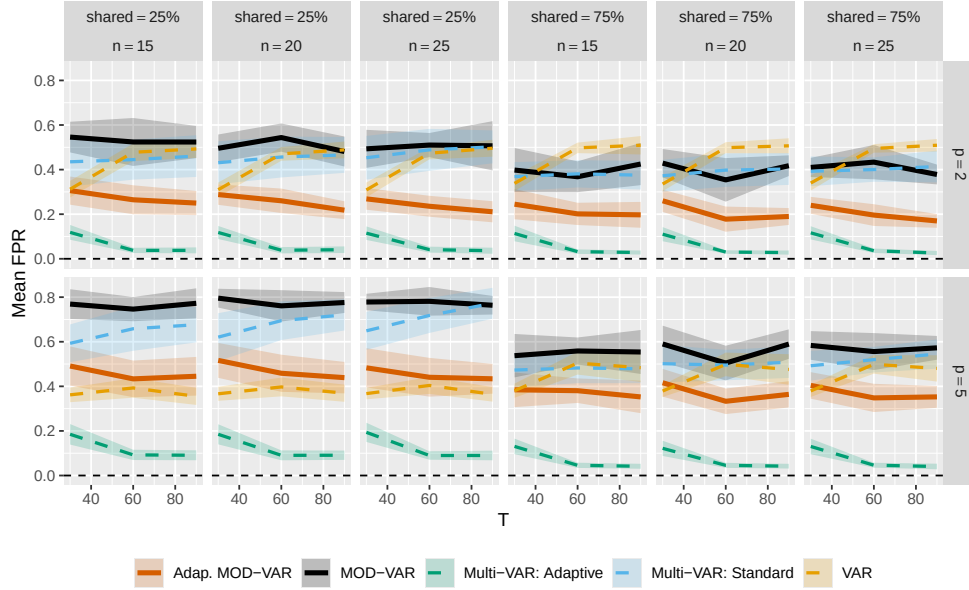


Figure S8: Visualization of the mean false positive rate (FPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects consists of only moderator-dependent components, with no subject-specific components. We explore performance for $d = 20$, varying n , T , p and the proportion of shared/moderated components.

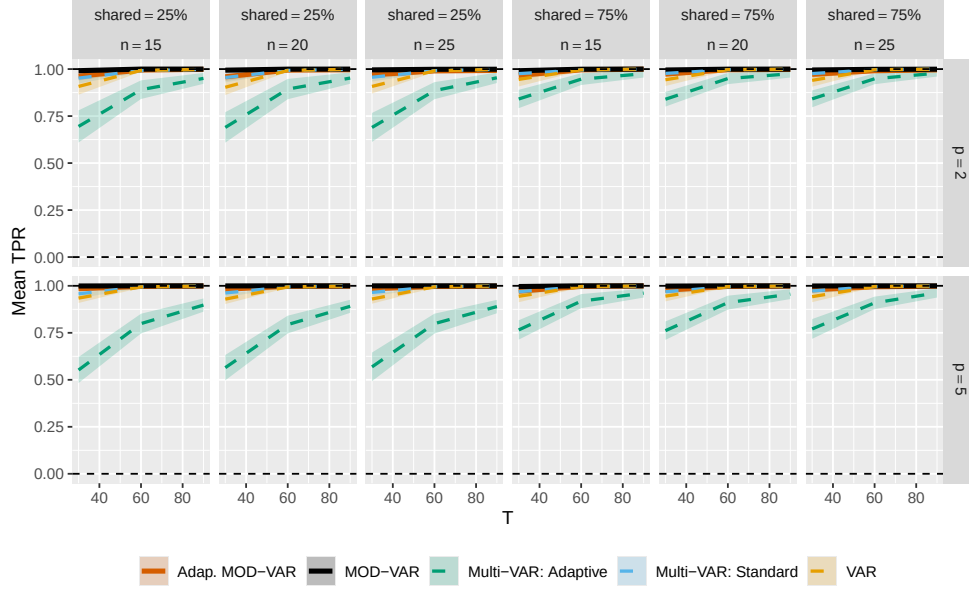


Figure S9: Visualization of the mean true positive rate (TPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects combines moderator-dependent and subject-specific components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

S10 and S12, for $d = 10, 20$, adaptive Multi-VAR and adaptive MOD-VAR attain the same FPR when $p = 5$ and 25% of entries are shared across subjects. Furthermore, as observed in Figures S9 and S11, the original and adaptive MOD-VAR attain visibly higher TPRs when compared to adaptive Multi-VAR. Thus, when $p = 5$ and 25% of entries are shared across subjects the MOD-VAR attains a clear advantage in terms of multi-subject GCN edge recovery. This helps further establish the adaptive MOD-VAR as a desirable method for recovering GCNs in a variety of numerical scenarios.

S2.3 Computational Time

We compare the computational time performance of our proposed MOD-VAR and other methods in the literature.

We visualize numerical computational time comparisons for data where heterogeneity across subjects consists of only moderator-dependent components for $d = 10$ in Figure S13. We observe that our original and adaptive MOD-VAR methods attain computational times comparable to the individual VAR model, and significantly lower than the Multi-VAR computational time for all simulation scenarios. This is because, when all heterogeneity

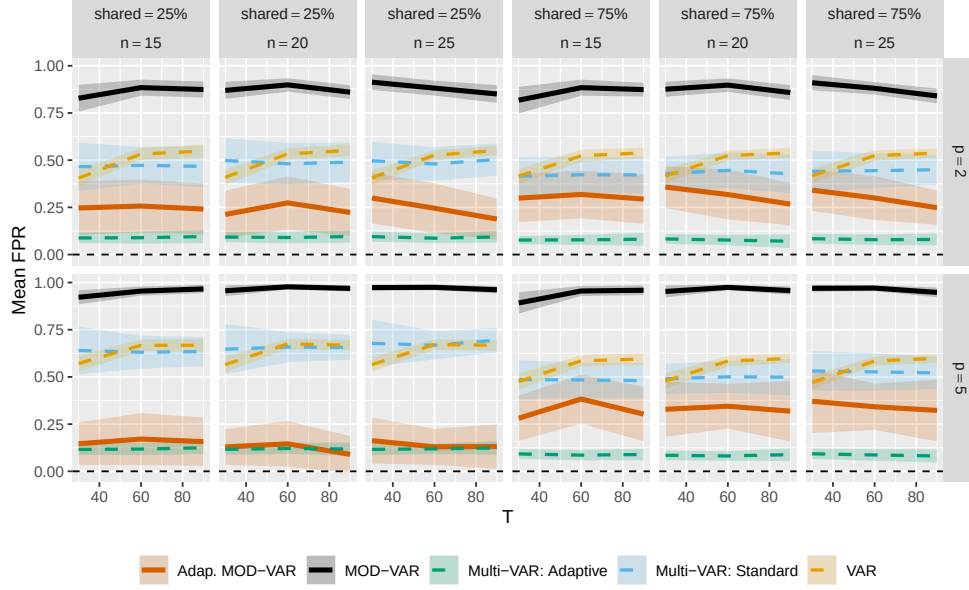


Figure S10: Visualization of the mean false positive rate (FPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects combines moderator-dependent and subject-specific components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderator-dependent components.

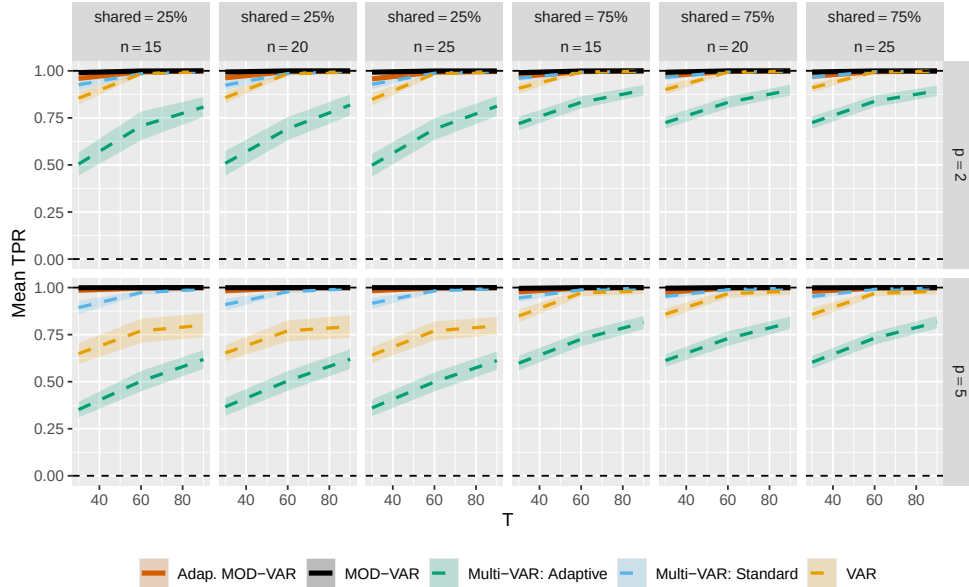


Figure S11: Visualization of the mean true positive rate (TPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects combines moderator-dependent and subject-specific components. We explore performance for $d = 20$, varying n , T , p and the proportion of shared/moderator-dependent components.

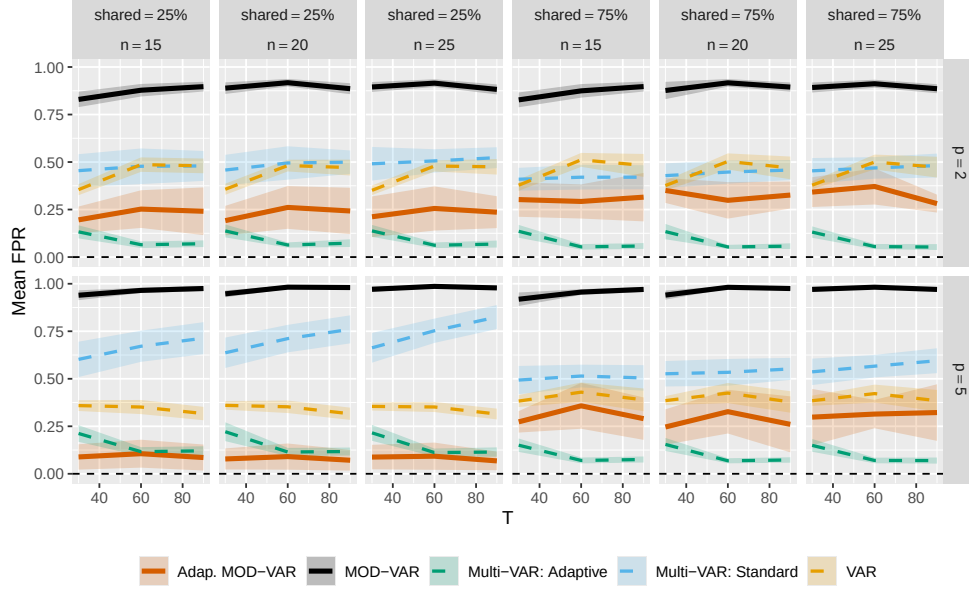


Figure S12: Visualization of the mean false positive rate (FPR) for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects combines moderator-dependent and subject-specific components. We explore performance for $d = 20$, varying n , T , p and the proportion of shared/moderator-dependent components.

is explained by additional moderator data, our proposed MOD-VAR attains a significant reduction in the number of estimated parameters in the model, which in turn reduces computational complexity of the model fitting procedure.

We provide additional numerical computational time comparisons for data where heterogeneity across subjects combines moderator-dependent and subject-specific components for $d = 10$ in Figure S14. By incorporating the estimation of subject-specific components, our proposed MOD-VAR must estimate a number of parameters comparable to the Multi-VAR procedure, increasing the computational time accordingly. A similar trend can be observed in the results comparing computational time in numerical experiments where heterogeneity consists of subject-specific components without moderator-dependent effects, as observed in Figure S15.

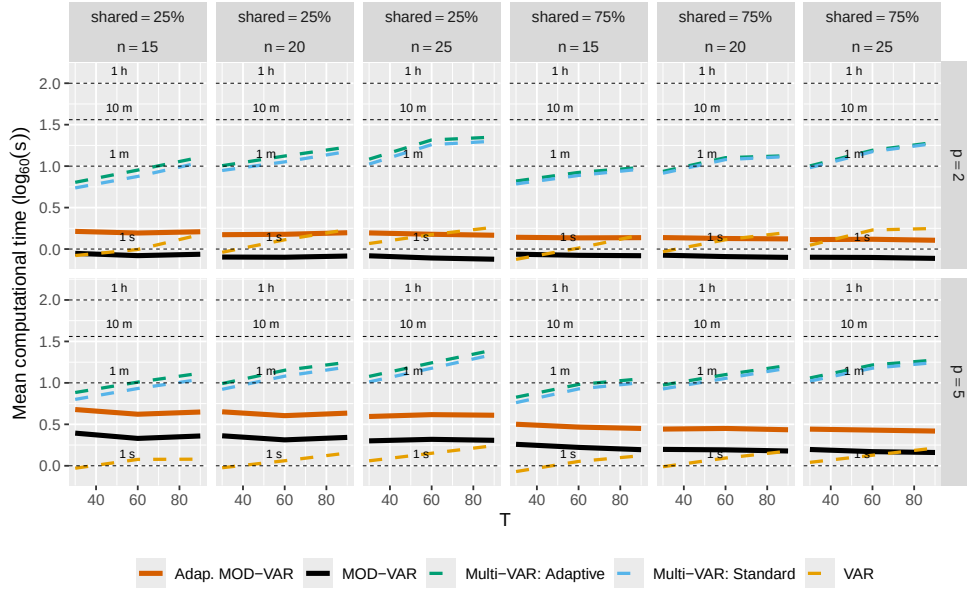


Figure S13: Visualization of the mean computational time for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects consists of only moderator-dependent components, with no subject-specific components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

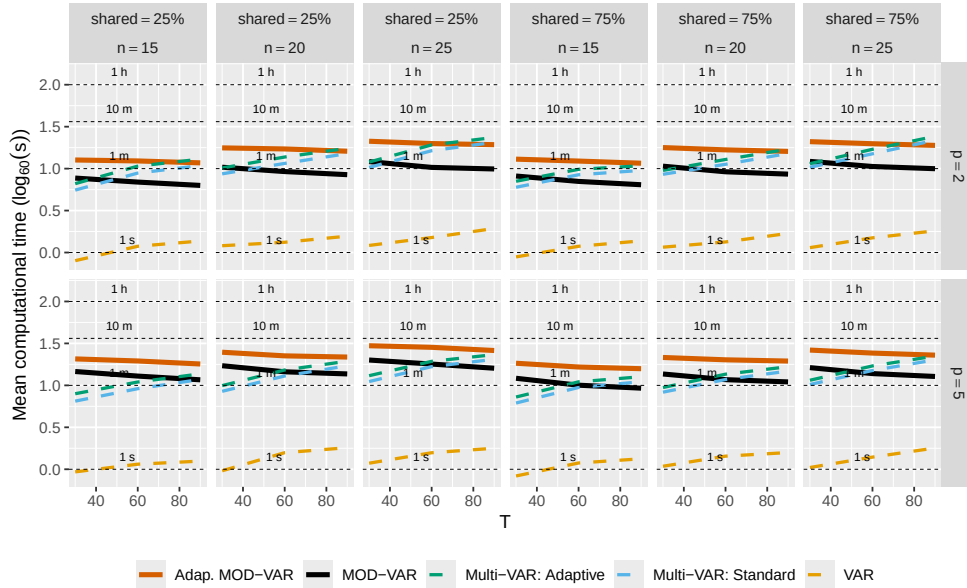


Figure S14: Visualization of the mean log-computational time for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects combines moderator-dependent and subject-specific components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

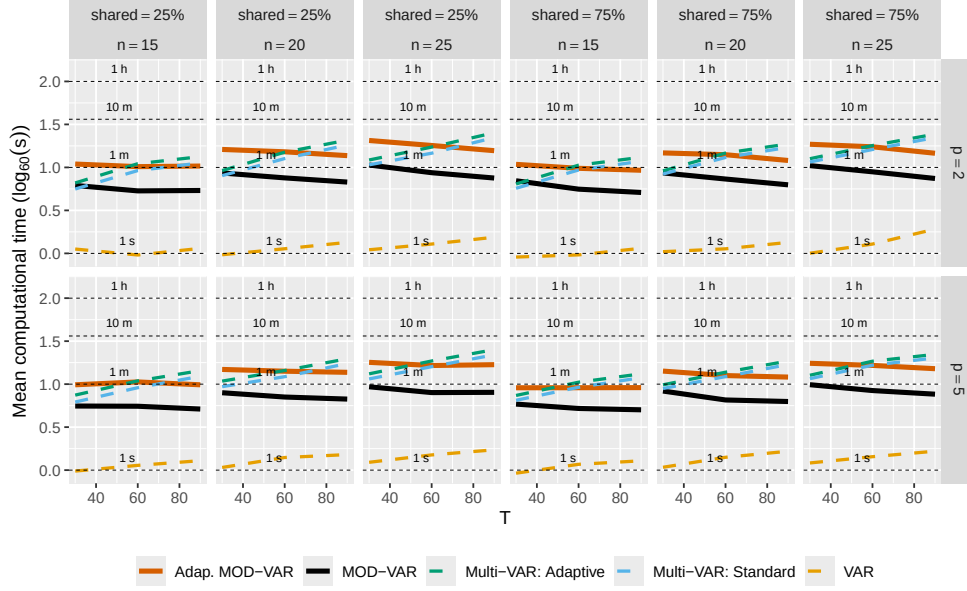


Figure S15: Visualization of the mean log-computational time for our proposed MOD-VAR and methods in the literature for data where the heterogeneity across subjects consists of only subject-specific components, without moderator-dependent components. We explore performance for $d = 10$, varying n , T , p and the proportion of shared/moderated components.

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